

Christopher Howell

Mechanical Design: Fusion 360 Parts & 3-Axis CAM

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SUMMARY

Simple mechanical parts, modeled right. I model mounts, brackets and rails as clean, parametric Fusion 360 solids and deliver them as STEP, STL and 3-axis CNC-ready files. I confirm critical dimensions before modeling, note which ones were confirmed or estimated, and keep communication plain. One person, start to finish.

EXPERIENCE

Howell Forge — Founder

Sylva, NC · 2023–present

- Model single parts from a sketch, a photo with dimensions, or an existing part as clean parametric Fusion 360 solids, delivered as STEP and STL.
- Prepare CNC-ready file sets: STEP, setup notes and a dimensioned PDF drawing.
- Build parametric variant packs: 3–5 sizes or hole-pattern variants of one base part, so later changes are quick.
- Confirm critical dimensions and hole patterns up front; every delivery includes notes on which dimensions were confirmed and which were estimated.
- Use AI-assisted tools in the CAD workflow to iterate faster and keep good notes. A person checks every dimension.
- Secondary lane: build AI business tools and apps, including GroundChiFlow (a mobile app) and Agent Forge.

SELECTED WORK

Picatinny / MIL-STD-1913 accessory rail — *modeled and cut*

Modeled in Fusion 360 and cut on a 3-axis machine. Tight, repeatable geometry with simple setups. Files: Fusion 360 model, STEP, NC program.

Prototype 1V8 — *bracket prototype, in revision*

Bracket modeled in Fusion 360 with Fusion CAM pocketing toolpaths and a simulation. CAM hygiene: multiple setups, organized NC programs, and the model sanity-checked against stock.

Bipod prototype — *Fusion 360 assembly, prototype*

Clamps, legs and adapter parts designed to fit together and be made on simple equipment. A prototype, not a product.

SKILLS

CAD	Fusion 360: parametric modeling, assemblies, dimensioned drawings
CAM	Fusion 360 CAM toolpaths and simulation; 3-axis CNC milling
Output	STEP, STL, NC programs
Tools	Python; AI-assisted workflow tooling (Cursor, local LLMs)

INTERESTS

Gear design for extreme sports: climbing, paragliding, kiteboarding and river sports.